Methylation-driven transcriptional silencing in clear-cell renal cell carcinoma: paired-sample integration of TCGA-KIRC DNA methylation and expression

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Abstract

Background. Paired tumor–adjacent-normal 450K analyses routinely declare more than half of tested CpGs differentially methylated, yielding lists whose biological actionability is limited. The mechanistically informative question is which methylation changes drive transcriptional silencing of their cognate gene — the epigenetic counterpart of Knudson's tumor-suppressor second hit. **Methods.** We integrate paired 450K DNA methylation and RNA-seq from TCGA-KIRC across 342 (patient, sample-type) keys (318 tumor + 24 adjacent-normal). For each of 137,451 promoter (CpG, gene) pairs we fit a linear mixed-effects regression of $\log_2(\mathrm{TPM}+1)$ on β with a patient random intercept and fixed-effect adjustment for sample type, age, sex, and tissue-source-site. A pair is declared *silencing* when a paired tumor-vs-normal moderated differential-methylation (DM) test and a within-status methylation–expression-association (MEA) test both reach FDR < 0.05 with concordant Knudson direction (hypermethylation in tumor and high β predicting low mRNA). **Results.** The screen yields **13,016 silencing pairs in 4,513 genes (9.5% of tested)** and recovers 13 of 18 predefined ccRCC tumor-suppressor biomarkers (hypergeometric 1.95×, $p = 0.014$). Silencing-restricted pathway enrichment recovers epithelial–mesenchymal transition, TGF-β signaling, axon guidance, and KEGG Pathways in Cancer; immune pathways that dominate raw-DM GSEA (Hallmark Inflammatory Response adj. $p = 6 \times 10^{-10}$) collapse to non-significance (adj. $p = 1.0$), consistent with the silencing filter automatically removing cell-composition artifacts without an external deconvolution panel. 82.8% of silencing events are subgroup-specific rather than cohort-wide; the canonical case VHL/`cg13672843` affects 32/318 tumors (10.1%) with a 33% mRNA reduction (Wilcoxon $p = 4.5 \times 10^{-6}$, $d = -0.79$). The pipeline replicates in TCGA-KIRP (papillary RCC). **Conclusion.** Restricting paired-design differential methylation to its transcription-coupled subset converts a pervasive but uninformative signature into a functionally interpretable, cell-composition-robust shortlist that recovers canonical cancer-driver loci.

1 Introduction

Paired tumor–adjacent-normal analyses on the Illumina 450K platform routinely declare ~2/3 of tested cytosine-guanine (CpG) positions differentially methylated at FDR < 0.05 — cancer epigenomes are pervasively disordered, with global hypomethylation of repetitive DNA and focal hypermethylation of CpG-island promoters (Feinberg & Vogelstein 1983; Jones 2012). The list is statistically valid but biologically uninformative: most differentially methylated CpGs are passenger or compositional changes, not regulators of transcription. The mechanistically interesting question is therefore not "which CpGs are differentially methylated" but **"which methylation changes drive transcriptional silencing of their cognate gene"** — the epigenetic counterpart of Knudson's tumor-suppressor second hit and the central paradigm of cancer epigenetics (Jones & Baylin 2007; Esteller 2008).

We address this in TCGA-KIRC, the largest paired-design cohort in TCGA, by integrating 450K β with RNA-seq $\log_2(\mathrm{TPM}+1)$ across 342 (patient, sample-type) keys with both modalities. For each promoter CpG we estimate the within-status association between β and the cognate gene's mRNA and require concordance with the differential-methylation direction before declaring transcriptional silencing. Clear-cell renal cell carcinoma is a natural setting: its molecular biology is dominated by VHL inactivation, and its single dominant histology limits biological heterogeneity (TCGA Research Network 2013; Kaelin 2007). Related methylation–expression screens (MethylMix, Gevaert 2015; ELMER, Yao et al. 2015; the eQTM literature) use correlation-based filtering; the present analysis differs by **(i)** prefiltering on paired-design DM rather than tumor-only DM, **(ii)** requiring the MEA direction to oppose the DM direction (Knudson signature), and **(iii)** treating the two together as a cell-composition control.

The framework adapts three statistical-genetics primitives to the omics setting: a paired moderated t-test for within-patient differential methylation, a dominant-encoded variant model repurposed as a per-CpG responder-fraction descriptor, and LD-based clumping repurposed as the one-CpG-per-gene representative used in pathway ranking. Multiple-testing control is by Benjamini–Hochberg FDR throughout, with a sign-flipping permutation null used to calibrate test inflation. The argument develops in five steps: paired-design DM on its own is pervasive but biologically diffuse (§3.1); transcriptional coupling shrinks the list by an order of magnitude to 13,016 silencing pairs (§3.2); directional concordance acts as a built-in cell-composition control, collapsing the immune-pathway block dominant in raw DM (§3.3); the dominant silencing pattern is patient-subgroup-specific rather than cohort-uniform, with VHL as the canonical illustration (§3.4); and the pipeline replicates in TCGA-KIRP with a biologically-interpretable biomarker-recovery pattern (§3.5–§3.6).

2 Methods

2.1 Cohort, data sources, and preprocessing

DNA methylation 450K β-values, RNA-seq STAR-Counts, and harmonized clinical metadata for TCGA-KIRC were retrieved via TCGAbiolinks (Colaprico et al. 2016). The methylation set comprises 484 aliquots from 319 patients (324 primary tumor and 160 solid adjacent-normal) after exclusion of one "additional new primary" aliquot; 160 patients contribute both tumor and adjacent-normal methylation aliquots and form the paired-methylation analysis set used for DM testing. The RNA-seq set comprises 614 aliquots, of which 605 unique (patient, sample-type) keys are retained; replicate aliquots are mean-aggregated. Matching the two modalities by (patient, sample-type) yields **342 keys (318 tumor + 24 adjacent-normal)** for the integration analysis. The 485,577-probe β-value matrix was sequentially filtered as in Table 1: cross-reactive and SNP-overlapping probes (MAF > 0.05 within 2 bp of the $3'$ end or the target CpG) were removed using DMRcate (Chen et al. 2013; Peters et al. 2015), probes with $> 20\%$ sample missingness were dropped, and complete-case rows were retained so every patient contributes both aliquots to every paired test. β-values were clipped to $[0.001, 0.999]$ and transformed to $M = \log_2(\beta/(1-\beta))$ for variance stabilization (Du et al. 2010). PCA on the top-10,000-variance CpGs ($n = 484$) shows PC1 (32.5% variance) cleanly separates tumor from adjacent-normal and PC2 (19.6%) tracks recorded sex (one-way ANOVA $R^2 = 0.94$); the latter motivates the chr X / chr Y exclusion above.

Table 1: Preprocessing attrition.

Step	Probes retained
Baseline (450K array)	485,577
Intersect with Illumina 450K annotation	472,651
Drop chr X / chr Y	461,295
Cross-reactive + SNP overlap (DMRcate)	433,595
Drop probes with >20% sample missingness	395,325
Complete-case across 484 samples	320,953

2.2 Differential methylation (DM) test

For each CpG, within-pair M-value differences $D_i = M_i^{\text{tumor}} - M_i^{\text{normal}}$ ($i = 1, \dots, 160$) were tested against H_0: $\mathbb{E}[D] = 0$ by a moderated paired t-test with empirical-Bayes shrinkage of the residual variance, implemented in limma (Smyth 2004; Ritchie et al. 2015). Patient-level covariates cancel in D_i. Benjamini–Hochberg FDR at $q < 0.05$ is used throughout the report unless otherwise stated.

2.3 Methylation–expression association (MEA) test

A promoter (CpG, gene) pair is built from the Illumina UCSC RefGene annotation: for each annotated CpG we expand to one row per gene–region entry, retain pairs where the region is `TSS1500`, `TSS200`, `5'UTR`, or `1stExon` — the four promoter-proximal regions defined by Illumina (within 1500 bp upstream of TSS, within

200 bp upstream of TSS, 5' untranslated region, and first exon respectively) — and de-duplicate so each (CpG, gene) pair appears once with the most-promoter-proximal label. Pairs are restricted to genes measured in RNA-seq, yielding 137,451 (CpG, gene) pairs across 128,023 unique CpGs and 15,842 unique genes.

For each pair we fit a linear mixed-effects model

$$\log_2(\mathrm{TPM}_{\mathrm{gene},i}+1) \;=\; \alpha+\beta\,\beta_{\mathrm{CpG},i}+\gamma^\top Z_i+u_{\mathrm{patient}(i)}+\varepsilon_i, \qquad Z_i=(\text{sample type, age, sex, tissue-source site})$$

across the 342 matched samples (318 unique patients; 24 contributing both tumor and adjacent-normal aliquots), with $u_{\mathrm{patient}} \sim N(0,\sigma_p^2)$ a random patient intercept that accounts for within-patient correlation among the 24 paired aliquots and $\varepsilon_i \sim N(0,\sigma^2)$ independent. Models are fit by REML in lme4 (Bates et al. 2015), with inference on β via Wald z-test from $\widehat{\mathrm{vcov}}(\hat{\beta})$. The sample-type covariate absorbs the tumor-vs-normal mean shift in both methylation and expression, so the test on β captures within-status methylation–expression association. The full screen of 137,451 pairs runs in ∼2 min on 16 cores; 2,156 pairs (1.57%) at near-invariant CpGs return degenerate fits and are dropped.

Two alternative formulations — pooled fixed-effects with QR partial-correlation residualisation, and a tumor-only refit on 318 independent tumors — yield near-identical results (supplementary); the mixed-effects specification is adopted as primary because it correctly models the paired structure.

The per-CpG **responder fraction** is $\hat{\pi}_{\mathrm{resp}} = \frac{1}{n_{\mathrm{tumor}}}\sum_{i\in\mathrm{tumor}} \mathbb{1}\{|\beta_i - \bar{\beta}_{\mathrm{normal}}| > 0.2\}$, the proportion of tumors whose β deviates from the adjacent-normal mean by more than 0.2 (a $\Delta\beta$ corresponding to a clear methylation transition). The threshold 0.2 is the standard biological significance cutoff in 450K analyses; the threshold for the responder fraction itself is discussed below. Conceptually, $\hat{\pi}_{\mathrm{resp}}$ corresponds to a **dominant-encoded** genetic-model treatment of methylation status — methylated above 0.2 / not-methylated — analogous to how an additive-vs-dominant variant encoding determines the design matrix in single-locus association tests.

2.4 Definition of silencing and pattern partition

A (CpG, gene) pair is **silencing** if it satisfies all three: (i) DM FDR $<$ 0.05, (ii) MEA FDR $<$ 0.05, and (iii) $\widehat{\mathrm{logFC}}_{\mathrm{DM}} > 0$ and $\hat{\beta}_{\mathrm{MEA}} < 0$ — that is, hypermethylation in tumor and high β predicting low mRNA. This is the strict Knudson-second-hit direction. The reverse direction (hypomethylation in tumor with high β predicting high mRNA) we label **activation** and report descriptively as a secondary observation; activation pairs are not the focus of this report.

Among silencing CpGs we further partition by responder fraction with a single descriptive cut at 0.30: **subgroup-driven silencing** ($\hat{\pi}_{\mathrm{resp}} < 0.3$, the rare-but-meaningful pattern exemplified by VHL) versus **cohort-wide silencing** ($\hat{\pi}_{\mathrm{resp}} \geq 0.3$, the SFRP1/RASSF1A-style uniform pattern). The 0.30 cut is operational, not a biological law; the responder fraction is best read as a continuous descriptor.

2.5 Pathway enrichment

Two pathway analyses are performed against MSigDB Hallmark (50 sets) and KEGG Legacy, including a KEGG Renal Cell Carcinoma set (Subramanian et al. 2005; Liberzon et al. 2015). (i) **GSEA**: each gene is ranked by $\mathrm{sign}(\hat{\beta}) \cdot -\log_{10}(p_{\mathrm{assoc}})$ at its best (lowest-p) promoter CpG, so that strongly silencing genes have strongly negative ranks; rankings are scored by the fast GSEA implementation of Korotkevich et al. (2021). Selecting one lead CpG per gene is the omics analog of LD clumping in GWAS: correlated promoter CpGs within a gene share local DNA-methylation structure, and collapsing to a single representative removes redundant signal in the gene-level ranking. (ii) **ORA**: the silencing gene set is the foreground, all 15,842 tested genes are the background, and pathway enrichment is tested by hypergeometric distribution. The raw-DM GSEA serves as the comparator to evaluate whether immune-pathway signals dominant in raw DM disappear under the silencing restriction.

2.6 Validation, calibration, and predefined controls

Predefined sanity checks: at least one VHL-promoter CpG must be flagged as silencing (positive control), and an 18-gene ccRCC literature TSG list (VHL, RASSF1A, SFRP1/2, GATA5, DKK3, …) is used as a benchmark. The DM-test genomic inflation $\lambda_{GC} = 36.8$ is calibrated against a sign-flipping permutation null on 30,000 probes × 20 replicates that gives $\lambda_{GC} = 1.0 \pm 0.4$, indicating real pervasive signal rather than miscalibration (Devlin & Roeder 1999).

3 Results

3.1 DM test: paired differential methylation is pervasive but uninformative on its own (Figure 1)

The DM test serves a dual role: it supplies one half of the silencing definition, and it is itself the de facto standard pipeline for paired-design 450K analyses — the benchmark against which the silencing filter's added selectivity must be measured. Of all 320,953 tested CpGs, **69% (222,386) reach FDR** < 0.05; the permutation null is flat ($\lambda_{GC}^{\text{perm}} \approx 1$), so the inflation reflects real pervasive signal rather than test miscalibration. Yet only 2.9% of CpGs clear the $|\Delta\beta| > 0.2$ biological-effect threshold (Figure 1). The two facts together are the motivating observation of this paper: paired-design DM is statistically valid but biologically diffuse, and statistical thresholding alone cannot recover a regulatorily interpretable shortlist.

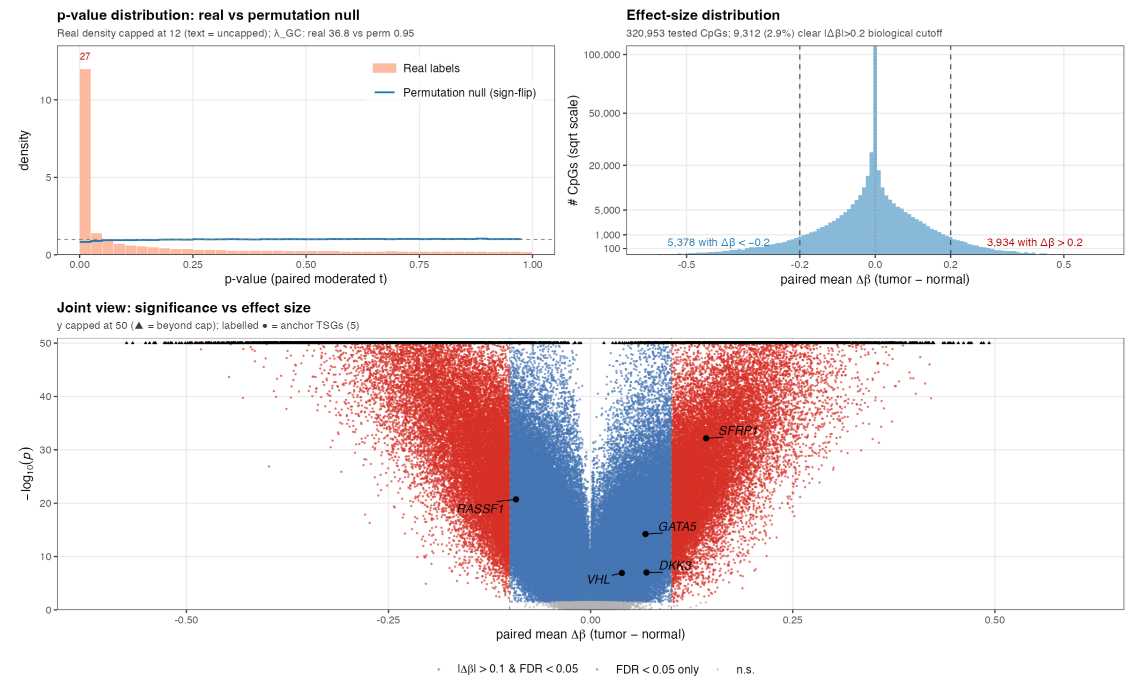

Figure 1: Paired tumor–normal DM diagnostic. Top-left: p-value distribution — real labels (peach; leftmost bin density 27, capped at 12 for plotting) vs sign-flip permutation null (blue, flat at ≈ 1); $\lambda_{GC}^{\text{real}} = 36.8$ vs $\lambda_{GC}^{\text{perm}} = 0.95$ visualizes that the inflation is real signal, not miscalibration. Top-right: paired-mean $\Delta\beta$ on a sqrt scale; 9,312 CpGs (2.9%) clear $|\Delta\beta| > 0.2$. Bottom: volcano on paired mean $\Delta\beta$ (y capped at 50; triangles = beyond cap) with five anchor TSGs — SFRP1/GATA5/DKK3 hypermethylated, VHL near origin (subgroup-driven, cohort-mean $\Delta\beta \approx 0$), RASSF1 hypomethylated (§4.2).

3.2 Methylation–expression integration identifies 13,016 silencing (CpG, gene) pairs (Figure 2)

The natural way to discipline the raw-DM list is to require each candidate CpG to actually predict its cognate gene's expression in the direction expected from gene silencing. Applying the integrated filter — paired DM, within-status MEA, and Knudson-direction concordance — to the 137,451 promoter (CpG, gene) pairs yields **13,016 silencing pairs (9.5% of tested) in 4,513 genes** (Figure 2, left). Fewer than one in ten promoter positions tested as differentially methylated turns out to be transcriptionally coupled in the silencing direction; the silencing list is roughly an order of magnitude smaller than the raw-DM list and, as the following sections show, has biology to support its size. Reverse-direction *activation* pairs are rarer still (2,247 pairs, silencing:activation $\approx 6:1$) and are reported descriptively only. The 9.5% rate is not driven by threshold

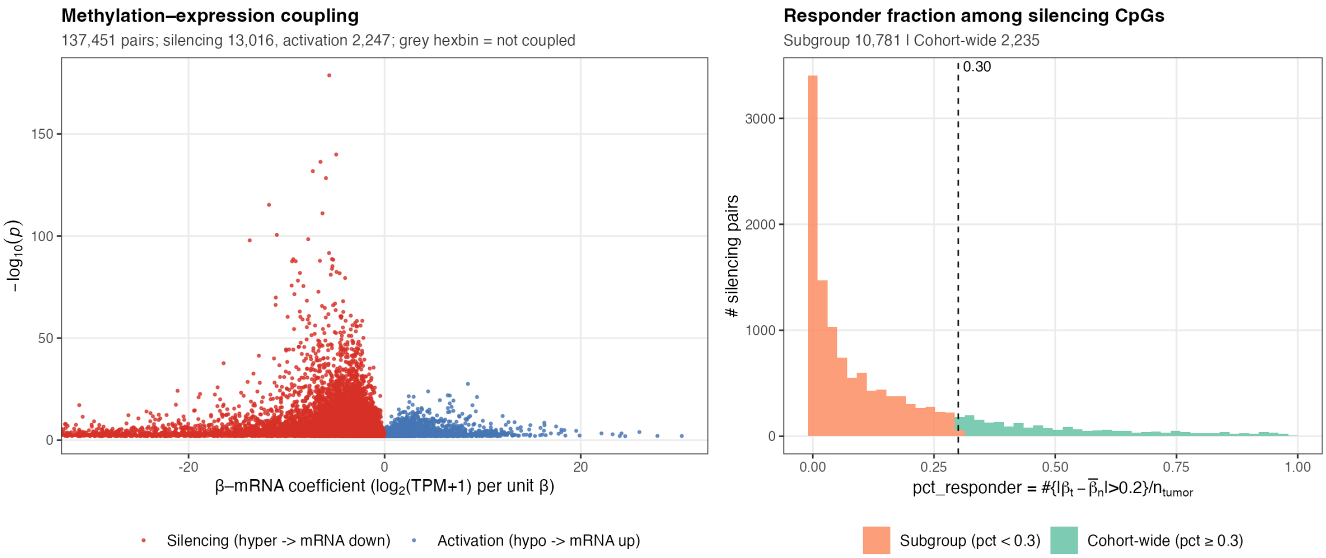

Figure 2: Methylation–expression coupling at promoter CpGs. Left: β–mRNA volcano; red = silencing (hypermethylation in tumor and within-cohort $\beta\uparrow\Rightarrow$ mRNA $\downarrow$); blue = activation (the reverse direction); grey = not coupled. Right: distribution of responder fraction among silencing pairs; most silencing (82.8%) is patient-subgroup-driven (orange, $\hat{\pi}_{\text{resp}} < 0.3$) rather than cohort-wide (green, ≥ 0.3).

choice: a 3×3 FDR grid varies the count within a factor of 1.75 but reproduces the same top-five pathways and the same subgroup-share rank order (supplementary).

3.3 Pathway enrichment: the silencing filter as a reference-panel-free cell-composition control (Figure 3)

Bulk tumor DM is confounded by cell composition: tumors carry more infiltrating leukocytes than adjacent normal kidney, so any CpG that distinguishes immune from epithelial cells appears differentially methylated for compositional reasons alone. The standard fix — reference-based deconvolution (EpiDISH, CIBERSORT) — adds a tool dependency and requires a tissue-matched reference panel. We hypothesise that silencing's directional concordance criterion automatically excises this signal: an immune-cell CpG bulk-hypomethylated in tumor *by* leukocyte infiltration also carries elevated bulk mRNA from the same infiltrating cells — the *opposite* of silencing's hypermethylation + low-mRNA signature — and therefore fails directional concordance and never enters the silencing list. The testable prediction is that pathways enriched in raw-DM purely on cell-composition grounds (notably immune pathways) should drop out of silencing-restricted enrichment.

The empirical contrast is striking. The raw-DM ranking is dominated by an eight-pathway immune block (Inflammatory Response, Complement, Allograft Rejection, Interferon-α/γ, TNFα–NF-κB, IL6–JAK–STAT3, Hematopoietic Cell Lineage) plus the KEGG Olfactory Transduction set — the latter a known GSEA artifact driven by ∼400 co-located olfactory receptors on 11p15. Under silencing-restricted ORA **all eight immune pathways collapse to** p_{adj} = 1.0 and the olfactory artifact vanishes (Table 2; Figure 3). What remains is recognisable cancer biology: EMT, TGF-β signaling, axon guidance (dominated by the protocadherin cluster, a documented ccRCC silencing target), ECM–receptor interaction, KRAS signaling, and the KEGG Pathways in Cancer set. The take-home is that two independent failure modes of bulk-tissue enrichment analysis — cell-composition confounding and set-size artifacts — are simultaneously neutralised by a filter whose only added structure is a per-CpG mixed-effects regression and a sign check.

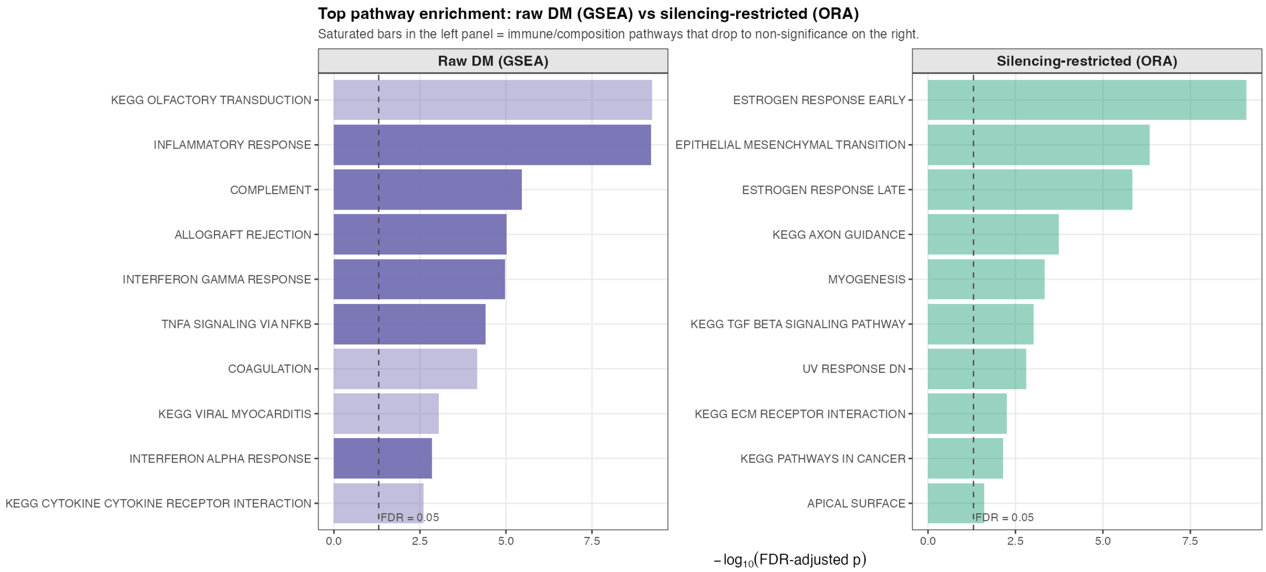

Figure 3: Top-10 pathway enrichment hits, raw-DM GSEA (left) vs silencing-restricted ORA (right). Raw-DM top hits are dominated by immune and coagulation sets (cell-composition-driven) plus KEGG Olfactory Transduction (a known large-family GSEA artifact); silencing-restricted top hits are cancer-relevant (EMT, TGF-β, axon guidance, KRAS, KEGG Pathways in Cancer). Left and right panels use different tests (signed GSEA vs unsigned hypergeometric ORA) because GSEA requires a per-gene ranking undefined for the silencing foreground; the comparison contrasts top-hit *content*, not test statistics.

Table 2: Immune-related pathways: collapse of significance from raw-DM GSEA to silencing-restricted ORA. k = silencing foreground genes in the set; K = total set size in the tested-gene background.

Pathway	Raw-DM GSEA padj	Raw-DM NES	Silencing ORA padj	k / K
Hallmark Inflammatory Response	6.0e-10	-2.21	1	31 / 185
Hallmark Complement	3.4e-06	-1.98	1	50 / 179
Hallmark Allograft Rejection	9.6e-06	-1.98	1	26 / 173
Hallmark Interferon-γ Response	1.1e-05	-1.93	1	34 / 182
Hallmark TNFα via NF-κB	4.0e-05	-1.85	1	40 / 189
Hallmark Interferon-α Response	1.4e-03	-1.83	1	15 / 88
KEGG Hematopoietic Cell Lineage	2.9e-03	-1.85	1	21 / 75
Hallmark IL6/JAK/STAT3 Signaling	2.5e-01	-1.40	1	16 / 78

3.4 Most silencing is patient-subgroup-specific; VHL/`cg13672843` is the canonical exemplar (Figures 2 right, 4, 5)

A separate question is whether silencing is a cohort-wide phenomenon or a patchwork in which different patient subsets silence different loci. The responder-fraction distribution is sharply right-skewed (Figure 2, right): **82.8% of silencing pairs are silenced in fewer than 30% of tumors**. The dominant pattern is therefore subgroup-specific, not cohort-uniform — with a direct methodological consequence. A cohort-mean DM test averages a minority responder signal against a non-responder majority and dilutes the effect toward zero, so a CpG can be biologically meaningful in a tenth of tumors yet near-invisible in a mean-difference test. Figure 4 illustrates the range across six anchor loci: SFRP1 carries the cohort-wide pattern; VHL, DKK3, GATA5, and PCDH17 each silence a different patient subset; RASSF1 is a ccRCC-specific hypomethylation anomaly addressed in §4.2.

VHL is the cleanest illustration of this dilution effect, and the analysis's predefined positive control (Figure 5). Its 1stExon CpG `cg13672843` carries a cohort-mean $\Delta\beta$ of only 0.04 — buried near the origin of the raw-DM volcano in Figure 1 — yet 32 tumors (10.1%, matching the documented ∼15% VHL methylation rate in ccRCC; Herman et al. 1994) carry $\beta \geq 0.2$, and these responder tumors show **a 33% reduction in VHL mRNA** relative to the non-responder majority. The naive cross-cohort linear correlation is essentially zero ($\rho = -0.03$) because it is dominated by the non-responder bulk; conditioning on sample type — the role the sample-type covariate plays in the MEA mixed model — recovers the within-status association at $p = 1.5\times10^{-6}$. VHL therefore demonstrates both why the silencing filter is necessary (mean tests miss it) and

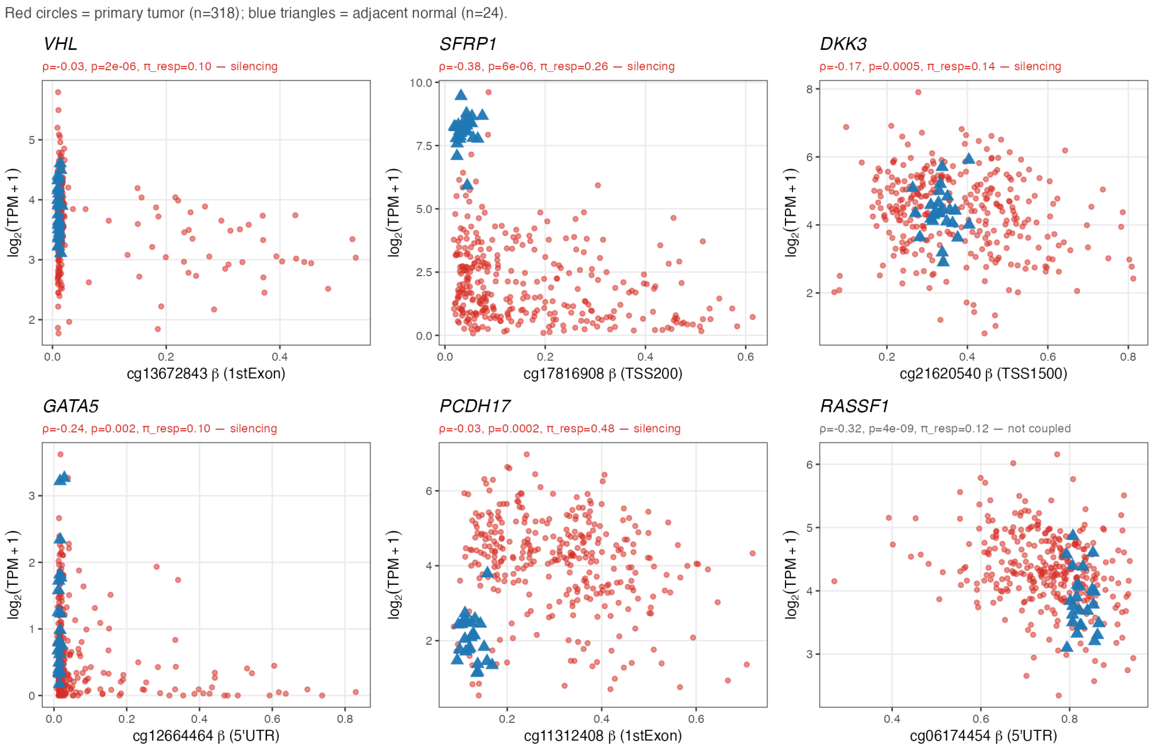

Figure 4: Promoter β vs gene expression at six representative loci. Each panel uses the per-gene best-association silencing CpG (or, for RASSF1, the best-association non-silencing CpG). VHL/cg13672843 = subgroup pattern; SFRP1, DKK3, GATA5, PCDH17 = silencing across patient subsets; RASSF1 = ccRCC-specific anomaly (§4.2).

why the within-status MEA specification is the right one (naive correlations mislead).

3.5 Literature biomarker recovery (Table 3)

The 18-gene panel of literature-reported ccRCC methylation-silenced biomarkers (VHL, RASSF1, SFRP1/2/4/5, DKK3, GATA5, BNC1, COL14A1, PCDH17, SLIT2, CDKN2A, APAF1, TIMP3, NEFH, UCHL1, PITX2) is a sanity check rather than a power-calibrated benchmark: it mixes extensively documented entries with single-cohort weak reports. With that caveat the panel partitions into three outcomes (Table 3): **10 genes recover as silencing** (the predefined positive control VHL, plus SFRP1/2, DKK3, GATA5, PCDH17, COL14A1, SLIT2, TIMP3, UCHL1); 3 carry an activation-direction CpG only (RASSF1, BNC1, SFRP5); and 5 are uncoupled (PITX2, APAF1, SFRP4, CDKN2A, NEFH). Against a 28.5% silencing-list baseline this is a hypergeometric enrichment of 1.95× ($p = 0.014$) — statistically real but modest, which is what one expects when the benchmark itself is heterogeneous. The substantive content is the within-panel pattern, not the overall percentage: VHL is recovered with the expected subgroup signature, the seven pan-RCC TSGs all recover as silencing, and the failures cluster on biomarkers whose primary inactivation

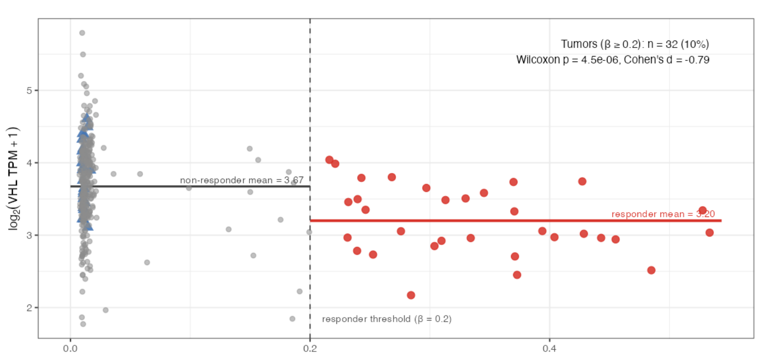

Figure 5: VHL/cg13672843 as a subgroup-driven silencing event. Red = responder tumors ($\beta \geq 0.2$, $n = 32$); grey = non-responder tumors ($n = 286$); blue = adjacent normals ($n = 24$). Responders show 33% lower VHL mRNA on the linear scale (Wilcoxon $p = 4.5 \times 10^{-6}$, Cohen's $d = -0.79$).

mechanism in ccRCC is not methylation (CDKN2A is dominated by deletion) or whose literature support is single-cohort.

3.6 External replication in TCGA-KIRP (papillary RCC)

A reproducibility test of a discovery pipeline is most informative when it is run unaltered on a distinct cohort. Applying the identical pipeline to 45 paired patients in TCGA-KIRP (papillary RCC) yields 9,116 silencing pairs in 2,955 genes, with **1,886 of those genes (64%) also silencing-recovered in KIRC** — a Jaccard of 0.34 against a hypergeometric expectation indistinguishable from zero. The pipeline therefore generalises beyond the cohort in which it was developed.

The cross-cohort pattern of the 18-gene biomarker panel is itself biologically informative. Five genes recover as silencing in both cohorts (SFRP1, DKK3, SLIT2, TIMP3, UCHL1 — the robust pan-RCC subset). Five recover only in KIRC, most importantly **VHL itself**, which is the predefined ccRCC-specific driver — an outcome consistent with the underlying biology rather than evidence of a fragile pipeline. Two recover only in KIRP, including **RASSF1**, which provides external evidence that the KIRC RASSF1 reversal (§4.2) is a tissue-specific anomaly and not a method failure. The remaining six biomarkers are silencing-recovered in neither cohort and overlap heavily with the §3.5 uncoupled and activation-only entries — i.e., where the literature support was weakest or the inactivation mechanism is not methylation. The apparent KIRP–KIRC list-size gap is a statistical-power artifact: re-anchoring to an n-independent effect-size threshold equalises the counts (KIRP/KIRC ratio 104%; supplementary).

Table 3: Silencing recovery for 18 predefined ccRCC methylation TSG biomarkers. # CpG = promoter CpGs tested; # sil / # act = silencing-direction / activation-direction CpGs passing the integration filter. RASSF1 is the conspicuous *activation-only* entry (hypomethylated rather than hypermethylated in KIRC); five genes are uncoupled, mostly reflecting deletion-driven or weakly-evidenced literature entries (§4.2, §3.6).

Gene	# CpG	# sil	# act	Status
RASSF1	37	0	1	Activation only
BNC1	18	0	1	Activation only
SFRP5	16	0	1	Activation only
PITX2	25	0	0	Not coupled
APAF1	21	0	0	Not coupled
SFRP4	11	0	0	Not coupled
CDKN2A	6	0	0	Not coupled
NEFH	6	0	0	Not coupled
TIMP3	13	2	1	Silencing recovered
UCHL1	3	3	0	Silencing recovered
SFRP2	34	4	0	Silencing recovered
VHL	7	1	0	Silencing recovered
SFRP1	15	5	0	Silencing recovered
SLIT2	6	3	0	Silencing recovered
COL14A1	11	2	0	Silencing recovered
PCDH17	13	4	0	Silencing recovered
DKK3	17	5	0	Silencing recovered
GATA5	19	6	0	Silencing recovered

4 Discussion

4.1 Differential methylation versus functional silencing

That two-thirds of paired-design CpGs reach FDR significance is not a pipeline failure: per-sample M-scale variance is small, sub-percent $\Delta\beta$ is detectable, and pervasive disordered methylation is the actual biology of cancer epigenomes. The corollary is that statistical thresholding cannot be the endpoint of a paired-design screen, and some criterion of regulatory consequence must take its place. Transcriptional coupling is the natural such criterion since promoter methylation's claim to gene-regulatory significance is precisely its ability to suppress transcription (Jones & Baylin 2007). The same logic explains, somewhat unexpectedly, why the silencing filter performs as a built-in cell-composition control. An immune-cell CpG that appears bulk-hypomethylated in tumor *because* the tumor contains more infiltrating leukocytes necessarily carries elevated bulk mRNA from those same infiltrating cells — the opposite of the Knudson signature — and is silently excluded by the directional concordance criterion. The collapse of the immune block in §3.3 is the empirical confirmation. The scope is narrow but worth being precise about: the silencing filter does not replace a deconvolution panel in general; it neutralises one specific confound (infiltrate-driven enrichment) by exploiting the asymmetry between methylation-driven and infiltration-driven directions.

4.2 Subgroup-driven silencing as the rule, not the exception

A second take-home is structural rather than methodological: most methylation-mediated silencing affects a minority of tumors, not the whole cohort (82.8% with $\hat{\pi}_{\text{resp}} < 0.3$ in KIRC, and a similar share in KIRP). The standard "common methylation silencing" framing implicitly assumes a cohort-wide signature; the data instead support an *alternative-routes-to-inactivation* view in which a tumor-suppressor gene can be lost through several mechanisms and methylation is the route taken in a tumor subset. VHL is the canonical example — methylated in $\sim$15% of ccRCC tumors despite gene loss in 60–80% (Kaelin 2007) — and the present screen suggests this is the rule rather than the exception. The methodological corollary is that cohort-mean differential testing systematically dilutes the dominant signal, and a per-tumor responder-fraction descriptor — conceptually a dominant-encoded variant model lifted from single-locus association — should be a routine complement.

4.3 Anomalies and benchmark heterogeneity

The most informative negative finding is RASSF1. Across its 37 promoter CpGs none pass strict silencing in KIRC, one passes the activation direction, and the best-CpG $\Delta\beta$ is -0.33 — i.e., RASSF1 is hypomethylated, not hypermethylated, in TCGA-KIRC, contrary to the canonical RASSF1A portrayal. The KIRP-recovers-but-KIRC-does-not pattern (§3.6) corroborates this as tissue-specific biology rather than a pipeline failure: the same filter applied to a histologically distinct RCC subtype reproduces the expected hypermethylation. The five biomarkers uncoupled in either cohort (PITX2, APAF1, SFRP4, CDKN2A, NEFH) are best read in the same light — CDKN2A is dominated by deletion/mutation in ccRCC, and the rest rely on single-cohort or otherwise weak literature support. The broader lesson is that a fixed literature panel is a useful sanity check on a discovery screen, but not a strict ground truth: the legitimate signal lives in the within-panel pattern (which biomarkers recover, which fail, and why), not in the overall recovery percentage.

4.4 Limitations

- **Bulk-tissue mixing** is reduced but not eliminated by the silencing filter; reference-based quantitative deconvolution was beyond course scope.
- **Single platform / single source**: KIRP external replication addresses single-cohort concerns within RCC but both cohorts are TCGA Illumina 450K — platform-independent validation (CPTAC, EPIC) is left to future work.
- **Association is not causation**: directional concordance with paired-design DM and the VHL mechanistic match are consistent with but do not prove causal silencing.
- **Promoter restriction**: gene-body and enhancer CpGs are excluded; non-promoter silencing is not captured.
- **Threshold dependence**: the $\hat{\pi}_{\text{resp}}$ and $\Delta\beta$ cuts are operational; KIRP replication uses identical parameters, and the cohort-size-independent effect-size sensitivity check (§3.6) shows the underlying signal is not threshold-driven.

5 Conclusion

Promoter methylation is informative as a gene-regulatory mechanism only insofar as it actually regulates a gene. Imposing that requirement on a paired-design TCGA-KIRC screen converts a pervasive but uninformative 222,386-CpG DM list into a 13,016-pair silencing shortlist that recovers canonical ccRCC drivers, neutralises the dominant immune-infiltrate confound without an external deconvolution panel, and reveals that most methylation-mediated silencing is patient-subgroup-specific rather than cohort-wide — a pattern that mean-based screens systematically dilute. The pipeline replicates in TCGA-KIRP, leaving platform-independent validation as the obvious next step.

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